

3.5 GRADIENT ECHO (GRE) & T2* WEIGHTING

MAXIMIZING SPEED AND SENSITIVITY TO SUSCEPTIBILITY

Gradient Echo (GRE) uses a small flip angle (α) and no 180° refocusing pulse. The signal comes from dephasing caused by magnetic field inhomogeneities, so GRE is fast and highly sensitive to T2* effects.

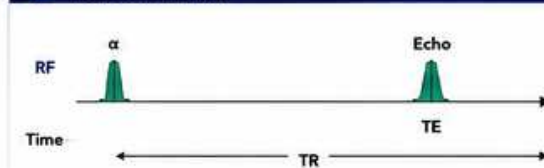


KEY CONCEPT

- No 180° pulse → faster, lower SAR, T1-weighted by default.
- T2* = T2 plus effects from field inhomogeneity.
- Shorter TE increases T2* weighting and reveals susceptibility.
- GRE family includes spoiled GRE (SPGR), gradient recalled echo (GRE), and multi-echo GRE.

- ↑ Increase / Benefit
↓ Decrease / Trade-off
— No / Minimal Effect

1 HOW GRE WORKS



- Small flip angle (α) tips magnetization into transverse plane.
- Gradients dephase spins → echo forms at TE (no 180° pulse).
- Signal decays with T2* (includes true T2 + B₀ inhomogeneity).
- TR sets T1 weighting; TE sets T2* weighting.

★ Faster than SE/FSE → shorter TR/TE and 3D capability.

2 GRE VARIANTS

Variant	Key Feature	Main Use
Spoiled GRE (SPGR)	RF spoiling (or gradient spoiling) destroys residual transverse magnetization.	T1-weighted 3D imaging (pre-contrast, anatomy)
Gradient Recalled Echo (GRE)	No refocusing pulse; coherent transverse magnetization.	T2*, functional, flow, balanced SSFP-like applications
Multi-echo GRE	Multiple echoes per TR (TE1, TE2, TE3 ...)	T2* mapping, iron quantification, susceptibility
3D GRE	Thin slices, often with spoiling and partition encoding.	High-resolution 3D anatomy (e.g., MPRAGE, VIBE, FSPGR)

3 NORMAL IMAGE APPEARANCE (1.5T BRAIN EXAMPLES)

T1-weighted GRE (SPGR) Short TE	T2*-weighted GRE Longer TE	SWI (a type of GRE) Very Long TE
• White matter brighter than gray • Good anatomic detail • Short TE (2–5 ms)	• More T2* contrast • Veins appear dark • Iron/calcium more conspicuous • TE ~ 15–30 ms	• Extremely T2* sensitive • Best for hemorrhage, microbleeds, calcifications • TE ~ 20–40+ ms

① Appearance depends on TR, TE, flip angle, coil, and field strength.

4 HOW PARAMETERS AFFECT GRE IMAGES

Parameter	Increase	Decrease	Effect
Flip Angle (α)	More T1 weighting, higher signal (to a point)	Less T1 weighting, lower signal	Controls T1 contrast and SNR
TR	Less T1 weighting, more T2* weighting	More T1 weighting, higher SNR	Short TR → faster scans, lower SNR
TE	More T2* weighting, more susceptibility	Less T2* weighting, more T1	Longer TE → more susceptibility effects
Bandwidth	Less chemical shift artifacts, lower SNR	More chemical shift artifacts, higher SNR	Higher BW → less distortion
Voxel Size	Less partial volume, lower SNR	More partial volume, higher SNR	Smaller voxels → higher resolution
Spoiling	Reduces coherence artifacts	Poor spoiling → banding, ghosting	Essential for T1 contrast and 3D

5 CLINICAL APPLICATIONS

	T1-weighted 3D Anatomy (SPGR)	Pre-contrast brain, liver, kidney, pituitary, contrast-enhanced imaging.
	SWI / Iron Imaging	Microbleeds, cavernomas, calcifications, venous structures, iron deposition.
	Cardiac Imaging	Cine, late gadolinium enhancement (LGE), perfusion, flow-sensitive applications.
	Angiography	Time-of-flight (TOF) MRA, contrast-enhanced MRA (3D GRE).
	Body Imaging	Liver dynamic studies, kidneys, pancreas, whole-body MR with short scan times.
	Lung Imaging	Ultrashort TE GRE captures signal in lungs where SE sequences fail.

★ GRE is the workhorse for speed, 3D imaging, and susceptibility-sensitive studies.

6 T2* EFFECTS AND WHAT THEY REVEAL

Less T2* Effect (Short TE)	Increasing T2* Effect (Longer TE)			Strong T2* Effect (Very Long TE)
Normal White Matter	Normal Veins	Iron Deposition	Calcium	Microbleeds / Hemorrhage
Relatively bright	Begin to appear dark	Dark signal	Very dark	Marked blooming (susceptibility)

① T2* decay is caused by microscopic field variations from deoxyhemoglobin, iron, calcium, metal, air-tissue interfaces, and susceptibility differences.

7 COMMON ARTIFACTS WITH GRE

Artifact	Cause	How to Reduce / Fix
Susceptibility (Blooming)	Field inhomogeneity from iron, air, metal, vessels	Shorter TE, higher bandwidth, advanced shimming, consider STIR/SE for comparison
Chemical Shift	Fat-water frequency difference	Higher bandwidth, correct frequency (CF), smaller voxel
Banding (SPGR)	Off-resonance / B ₀ inhomogeneity	Prescan, correct CF, shim, higher bandwidth
Ghosting	Motion, flow, incorrect spoiling	Patient immobilization, flow compensation, proper spoiling settings
Off-center Signal Loss	B1 inhomogeneity, short TR, low flip	Adjust flip angle, optimize TR, use dielectric pads if available

8 ENGINEER CHECKLIST (GRE)

☒ Is prescan center and shim adequate?	☒ Is RF/gradient spoiling enabled (SPGR)?
☒ Is center frequency (CF) correct (no banding)?	☒ Are coils properly connected and tuned?
☒ Is flip angle appropriate for goal (T1 vs SNR)?	☒ Are gradients and RF systems performing within specs?
☒ Is TR/TE set for the desired contrast?	☒ Any metal / susceptibility sources in or near FOV?
☒ Is bandwidth high enough to minimize artifacts?	☒ Repeat test or sequence to confirm if needed.

QUICK PARAMETER IMPACT (GRE)

Parameter	Change	Result
↑ Flip Angle	↑	More T1 weighting, ↑ signal (to optimum)
↑ TR	↓	More T2* weighting, ↓ scan time
↑ TE	↑	More T2* weighting, ↑ susceptibility
↑ Bandwidth	↓	Less artifact, ↓ SNR
↑ Spoiling	↓	Better T1 contrast, fewer banding/ghosts

9 PRACTICAL TIPS

- Start with short TE for anatomy; lengthen TE to increase susceptibility when needed.
- For T1 3D imaging, use moderate flip angle and spoiling for best contrast.
- Always verify CF and shim before long GRE acquisitions.
- Compare with SE or FSE if artifact vs pathology is uncertain.



REMEMBER

GRE is sensitive to everything that distorts the magnetic field. That sensitivity is powerful — and also the source of most artifacts.



KEY TAKEAWAY

GRE = speed + T1 by default + T2* sensitivity. Understand TR, TE, flip angle, and spoilers to get the right contrast and manage artifacts.